

MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
TECHNICAL SERVICES & INSPECTION DEPARTMENT
NON-DESTRUCTIVE TESTING (NDT) EXAMINATION REPORT

REPORT REF: MRPL/NDT/CDU-01/2026/0488
DATE OF INSPECTION: 14-JAN-2026
PLANT / COMPLEX: Phase-1 Crude Distillation Unit (CDU-1)
EQUIPMENT TAG: CDU-Col-04 (Atmospheric Distillation Column)
SERVICE: High-Sulfur Middle East Crude Distillation (Feed: Arab Heavy Blend)
OPERATING TEMPERATURE: 384 deg C | OPERATING PRESSURE: 4.8 bar gauge
INSPECTION METHOD: High-Temperature Ultrasonic Pulse-Echo Contact Testing (UT)
TEST PROCEDURE: MRPL-SOP-NDT-UT-04 / ASME Section V, Article 5
TEST EQUIPMENT: Olympus Epoch 650 with Dual Element Probe D790-SM (5 MHz)
CALIBRATION STANDARD: IIW Type 1 Carbon Steel Calibration Block (Step Block 2-20mm)

EXECUTIVE INSPECTION SUMMARY & API 510 COMPLIANCE EVALUATION

1. INSPECTION OBJECTIVE:

Pre-turnaround thickness survey of Column-04 bottom shell course (Elev +4.0m to +7.0m) to determine remaining wall thickness and calculate localized corrosion rates resulting from processing high-TAN sour crude blends over the preceding 36-month operating campaign.

2. DESIGN PARAMETERS (NAMEPLATE DATA):

- Original Nominal Shell Thickness: 14.0 mm
- Shell Material of Construction: ASTM A516 Grade 70 (Killed Carbon Steel)
- Original Corrosion Allowance: 3.0 mm
- API 510 Minimum Retirement Thickness (t_{min}): 6.0 mm (calculated per UG-27)

3. ULTRASONIC THICKNESS MEASUREMENT GRID (CML READINGS):

Grid Point	Elevation	Orientation	Nominal	Actual t	Deficit	Status
CML 01-A	Elev +4.2m	North (0 deg)	14.0 mm	4.20 mm	-1.80 mm	CRITICAL RETIREMENT
CML 01-B	Elev +4.2m	East (90 deg)	14.0 mm	4.35 mm	-1.65 mm	CRITICAL RETIREMENT
CML 01-C	Elev +4.2m	South(180 deg)	14.0 mm	4.18 mm	-1.82 mm	CRITICAL RETIREMENT
CML 01-D	Elev +4.2m	West (270 deg)	14.0 mm	4.25 mm	-1.75 mm	CRITICAL RETIREMENT
CML 02-A	Elev +4.8m	North (0 deg)	14.0 mm	4.62 mm	-1.38 mm	CRITICAL RETIREMENT
CML 02-B	Elev +4.8m	East (90 deg)	14.0 mm	4.70 mm	-1.30 mm	CRITICAL RETIREMENT
CML 02-C	Elev +4.8m	South(180 deg)	14.0 mm	4.58 mm	-1.42 mm	CRITICAL RETIREMENT
CML 02-D	Elev +4.8m	West (270 deg)	14.0 mm	4.65 mm	-1.35 mm	CRITICAL RETIREMENT
CML 03-A	Elev +5.5m	North (0 deg)	14.0 mm	7.10 mm	+1.10 mm	HIGH CORROSION
CML 03-B	Elev +5.5m	East (90 deg)	14.0 mm	7.25 mm	+1.25 mm	HIGH CORROSION
CML 03-C	Elev +5.5m	South(180 deg)	14.0 mm	6.95 mm	+0.95 mm	MONITOR CLOSELY
CML 03-D	Elev +5.5m	West (270 deg)	14.0 mm	7.05 mm	+1.05 mm	HIGH CORROSION

4. CORROSION RATE CALCULATION:

- Previous UT reading (Turnaround 2023): 6.80 mm at CML 01-C
- Current UT reading (January 2026): 4.18 mm at CML 01-C
- Operating interval: 3.0 years
- Short-term Corrosion Rate: $(6.80\text{ mm} - 4.18\text{ mm}) / 3.0\text{ years} = 0.873\text{ mm/year!}$
- Remaining Life Projection: $(4.18\text{ mm} - 6.00\text{ mm}) / 0.873 = \text{NEGATIVE (EXPIRED)}$

5. INSPECTOR RECOMMENDATION & MANDATORY ACTION:

In accordance with API 510 Section 7 and statutory factory inspectorate norms (OISD-STD-129), CDU Column-04 bottom section cannot safely operate past the current cycle without structural reinforcement. Immediate emergency shutdown and in-situ weld overlay cladding using Inconel 625 alloy (min 3.0 mm thickness) or replacement of bottom shell ring course is mandatory.

Inspected By: S. N. Rao, Senior Inspection Engineer (ASNT Level III UT / API 510 #48291)
Approved By: Chief Manager (Technical Services & Asset Integrity), MRPL Mangalore